

DSA LABSHEET-8

AM.SC.U4CSE25129

K.ROHITH

1 and 2)

```
package labsheet9;
import java.util.Scanner;
class BinaryTree {
    Scanner sc = new Scanner(System.in);
    Node createTree() {
        System.out.print("Enter data (-1 for no node): ");
        int data = sc.nextInt();
        if (data == -1) {
            return null;
        }
        Node newNode = new Node(data);
        System.out.println("Enter left child of " + data);
        newNode.left = createTree();
        System.out.println("Enter right child of " + data);
        newNode.right = createTree();
        return newNode;
    }

    boolean isEmpty(Node root) {
        return root == null;
    }

    void display(Node root, int space) {
        if (root == null) {
            return;
        }
        space = space + 5;
        display(root.right, space);
        System.out.println();
        for (int i = 5; i < space; i++) {
            System.out.print(" ");
        }
        System.out.println(root.data);
        display(root.left, space);
    }

    void inorder(Node root) {
        if (root == null) {
            return;
        }
        inorder(root.left);
        System.out.print(root.data + " ");
        inorder(root.right);
    }

    void preorder(Node root) {
        if (root == null) {
            return;
        }
    }
```

```

    }
    System.out.print(root.data + " ");
    preorder(root.left);
    preorder(root.right);
}
void postorder(Node root) {
    if (root == null) {
        return;
    }

    postorder(root.left);
    postorder(root.right);
    System.out.print(root.data + " ");
}
}

```

```

1 package dsalabsheet8;
2
3 public class Main {
4
5     public static void main(String[] args) {
6
7         BinaryTree tree = new BinaryTree();
8         Node root = tree.createTree();
9         if(root==null) {
10             System.out.println("\nTree is Empty");
11         }else {
12             tree.display(root, 0);
13         }
14     }
15 }
16
17 }
18

```

```

1 package dsalabsheet8;
2
3 class Node {
4     int data;
5     Node left, right;
6     Node(int data) {
7         this.data = data;
8         left = right = null;
9     }
10 }
11

```

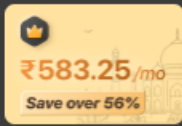
```
Enter data (-1 for no node): 4
Enter left child of 4
Enter data (-1 for no node): 3
Enter left child of 3
Enter data (-1 for no node): 2
Enter left child of 2
Enter data (-1 for no node): -1
Enter right child of 2
Enter data (-1 for no node): -1
Enter right child of 3
Enter data (-1 for no node): 1
Enter left child of 1
Enter data (-1 for no node): -1
Enter right child of 1
Enter data (-1 for no node): -1
Enter right child of 4
Enter data (-1 for no node): 5
Enter left child of 5
Enter data (-1 for no node): 6
Enter left child of 6
Enter data (-1 for no node): -1
Enter right child of 6
Enter data (-1 for no node): -1
Enter right child of 5
Enter data (-1 for no node): 7
Enter left child of 7
Enter data (-1 for no node): -1
Enter right child of 7
Enter data (-1 for no node): -1
Tree is not empty.
```

5)

```
16 class Solution {
17     public List<List<Integer>> levelOrder(TreeNode root) {
18         List<List<Integer>> result = new ArrayList<>();
19         Queue<TreeNode> q = new LinkedList<>();
20         if(root==null){
21             return result;
22         }
23         q.add(root);
24         while(!q.isEmpty()){
25             int size = q.size();
26             List<Integer> level = new ArrayList<>();
27             for(int i=0;i<size;i++){
28                 TreeNode current = q.remove();
29                 level.add(current.val);
30                 if(current.left!=null){
31                     q.add(current.left);
32                 }
33                 if(current.right!=null){
34                     q.add(current.right);
35                 }
36             }
37             result.add(level);
38         }
39         return result;
40     }
41 }
```

Accepted 35 / 35 testcases passed Time taken: 1d 6hrs 59m 46s

 rohith submitted at Oct 05, 2026 17:49



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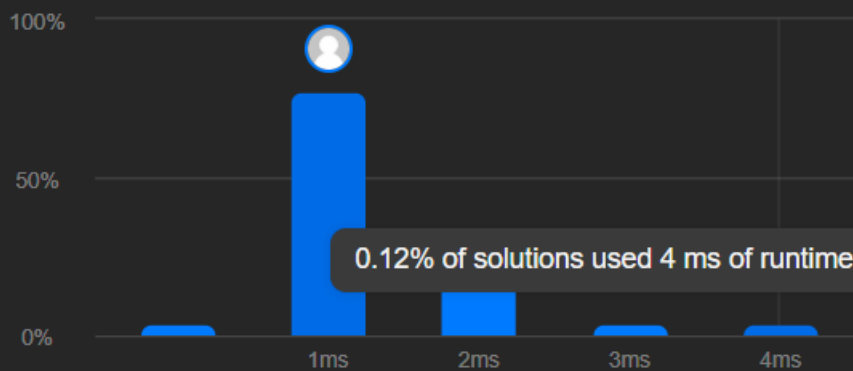


 Runtime

1 ms | Beats **96.27%** 

 Memory

46.90 MB | Beats **22.19%**



0.12% of solutions used 4 ms of runtime

1ms

2ms

3ms

4ms